

Name: _____ Register No: _____

Experiment 12

Question: Experiment to detect outliers in a given data set.

Aim

To detect outliers in a given data set using the Z-score and IQR methods.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_12():
    rng = np.random.default_rng(4)
    salary = np.append(rng.normal(50000, 8000, 100).round(0), [150000, 5000, 180000])
    df = pd.DataFrame({"Salary": salary})
    print(df["Salary"].describe().round(2), "\n")

    # Method 1: Z-score (|z| > 3)
    z = (df["Salary"] - df["Salary"].mean()) / df["Salary"].std()
    out_z = df[np.abs(z) > 3]
    print("Z-score method (|z| > 3):", len(out_z), "outliers")
    print(out_z["Salary"].tolist(), "\n")

    # Method 2: IQR (below Q1-1.5*IQR or above Q3+1.5*IQR)
    q1, q3 = df["Salary"].quantile([0.25, 0.75])
    iqr = q3 - q1
    low, high = q1 - 1.5 * iqr, q3 + 1.5 * iqr
    out_iqr = df[(df["Salary"] < low) | (df["Salary"] > high)]
    print(f"IQR method: Q1={q1:.0f}, Q3={q3:.0f}, IQR={iqr:.0f}")
    print(f"Lower limit={low:.0f}, Upper limit={high:.0f}")
    print(len(out_iqr), "outliers:", sorted(out_iqr["Salary"].tolist()), "\n")

    # Removing outliers
    clean = df[(df["Salary"] >= low) & (df["Salary"] <= high)]
    print("Rows before:", len(df), "| Rows after removal:", len(clean))
    print("Mean before: %.2f | Mean after: %.2f" % (df["Salary"].mean(), clean["Salary"].mean()))

    fig, ax = plt.subplots(1, 3, figsize=(14, 4.5))
    sns.boxplot(y=df["Salary"], ax=ax[0], color="salmon").set_title("Box plot (with outliers)")
    ax[1].scatter(range(len(df)), df["Salary"], c=((df["Salary"] < low) | (df["Salary"] > high)),
                  cmap="coolwarm")
    ax[1].axhline(low, color="r", linestyle="--")
    ax[1].axhline(high, color="r", linestyle="--")
    ax[1].set_title("Scatter plot (red = outlier)")
    sns.boxplot(y=clean["Salary"], ax=ax[2], color="lightgreen").set_title("Box plot (after removal)")
    plt.tight_layout()
    plt.show()
```

exp_12()

Output

```
count      103.00
mean       51304.44
std        18611.48
min         5000.00
25%        43695.00
50%        49120.00
75%        56088.50
max       180000.00
Name: Salary, dtype: float64
```

```
Z-score method (|z| > 3): 2 outliers
[150000.0, 180000.0]
```

```
IQR method: Q1=43695, Q3=56088, IQR=12394
Lower limit=25105, Upper limit=74679
3 outliers: [5000.0, 150000.0, 180000.0]
Rows before: 103 | Rows after removal: 100
```

Mean before: 51304.44 | Mean after: 49493.57

Graph

